

Databricks (Free Edition) — Cloud Alternative to the Local Lab

The Week 6 PySpark/Delta notebook is designed to run **locally** in Docker (see `../pyspark/`) so no account is needed. This directory explains how to run the **same concepts** on the real Databricks platform instead, using the free tier.

Free Edition vs Community Edition: Databricks Free Edition (launched June 2025) replaced the legacy Community Edition, which was retired on **1 Jan 2026**. Free Edition is serverless and quota-limited — but needs **no cloud account and no credit card**.

1. Create a free workspace

1. Go to <https://www.databricks.com/learn/free-edition> (or <https://login.databricks.com>).
2. Sign up with Google, Microsoft, or an email address.
3. Databricks automatically provisions a **serverless workspace** and drops you into the UI. There is nothing to install.

2. The workspace at a glance

The left sidebar groups everything:

Area	What it holds
Workspace	Notebooks, folders, Git repos
Catalog	Unity Catalog: catalogs → schemas → tables & volumes
Compute	Clusters / serverless compute to attach notebooks to
SQL	SQL editor, queries, dashboards, SQL warehouses
Jobs / Pipelines	Scheduling & orchestration (Lakeflow)

A notebook **attaches to compute**, then runs Python / SQL / Scala / R cells.

3. Run the Week 6 notebook on Databricks

1. In the sidebar, **Workspace** → **(your home)** → **Import** and upload `code/week6_databricks_notebook.ipynb`.
2. Attach the notebook to **serverless compute** (top of the notebook).
3. **Delete (or skip) the SparkSession setup cell.** On Databricks a `spark` session already exists — you do *not* call `configure_spark_with_delta_pip`.
4. Change the storage path. The local notebook writes to `/home/jovyan/work/delta`. On Databricks, either:
 - write to a **Unity Catalog volume**, e.g. `/Volumes/<catalog>/<schema>/<volume>/delta`, or
 - use **managed tables**:
`df.write.format("delta").saveAsTable("workspace.default.bronze_orders")`

and read them back with `spark.read.table(...)`.

5. Everything else — `.filter`, `.groupBy().agg()`, the Bronze→Silver→Gold medallion flow, `DeltaTable.history()`, and `versionAsOf` time travel — runs **unchanged**, because the DataFrame and Delta APIs are identical.

4. (Optional) Drive Databricks from VS Code

Prefer your editor to the browser notebook UI:

1. Install the **Databricks extension for Visual Studio Code**.
2. Authenticate to your Free Edition workspace (the extension walks you through host + auth).
3. Enable **Databricks Connect** (works with Databricks Runtime 13.3+). Cells run locally while DataFrame operations execute on remote Databricks compute, and results stream back into the VS Code UI.

References

- Free Edition signup & overview: <https://docs.databricks.com/aws/en/getting-started/free-edition>
- VS Code extension + Databricks Connect: <https://docs.databricks.com/aws/en/dev-tools/vscode-ext/>
- Local (no-account) alternative: `./pyspark/readme.md`